

CONTENT

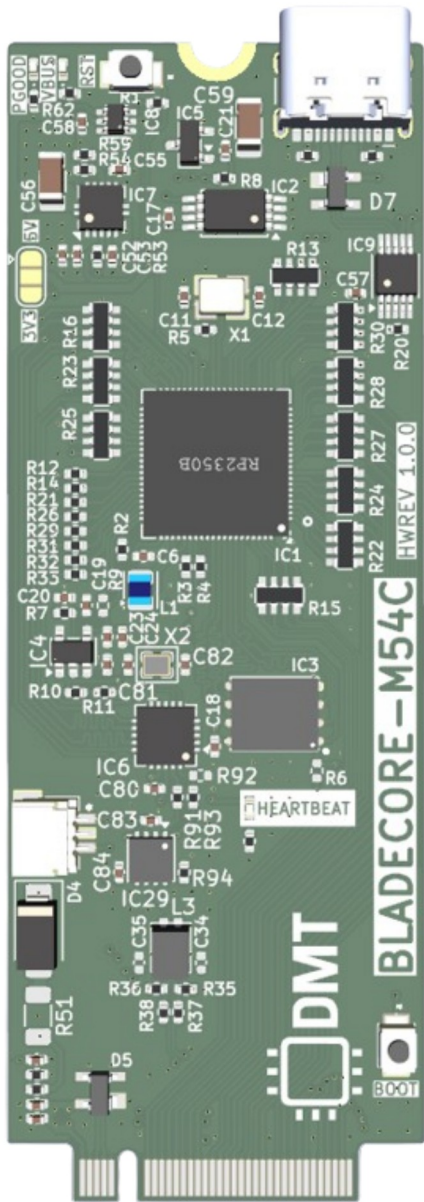
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COMMENT GUIDELINES

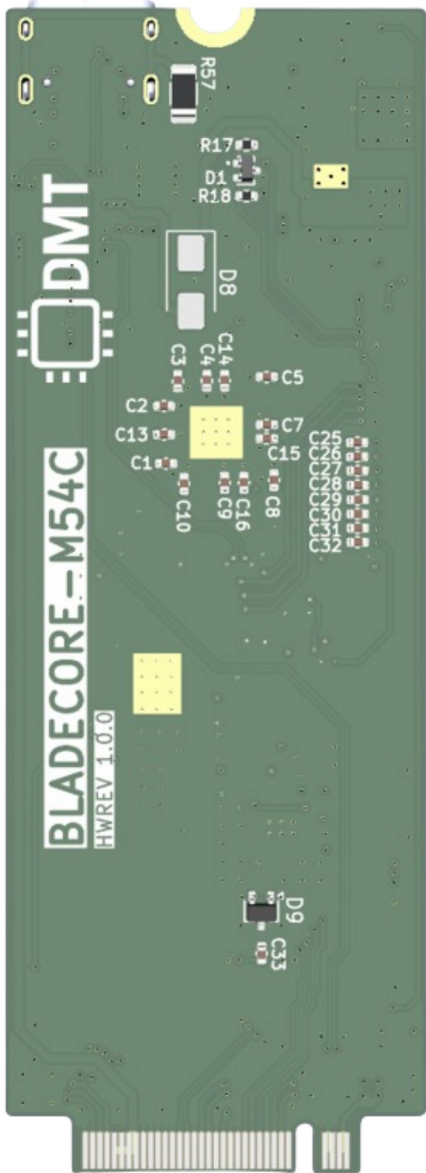
General comments are black, 50 mil size
Design notes and guidelines are blue, 50 mil size
Layout instructions are red, 50 mil size

PCB PREVIEW

TOP VIEW



BOTTOM VIEW



NOTES

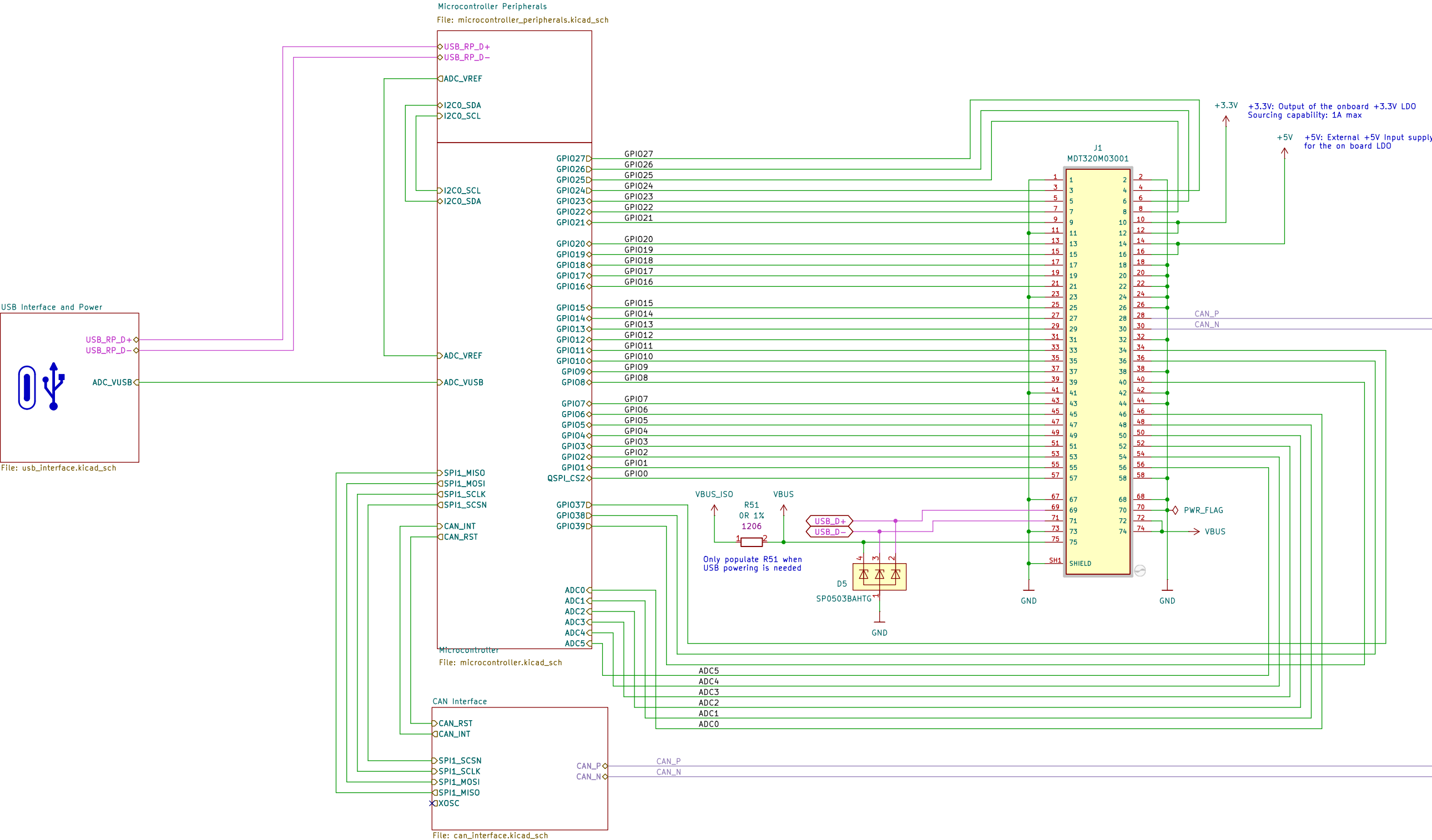
Not fitted components are marked as X

 Date: 2026-01-31 Sheet Title: Root	Board Name: BladeCore		Project Name: M54	
	File Name: BladeCore-M54C.kicad_sch		Revision: 1.0.0	Variant: C
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 1 of 9

Input voltage:	3.3V - 5.5 V
Absolut maximum current consumption	~200 mA
Typical current consumption	60mA
IO voltage	3.3V
Onboard LDO output capacity	1.5A
LDO Dropout	60mV

DMT Date: 2026-01-31	Board Name: BladeCore		Project Name: M54	
	File Name: Block Diagram.kicad_sch		Revision: 1.0.0	Variant: C
Sheet Title: Block Diagram	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 2 of 9

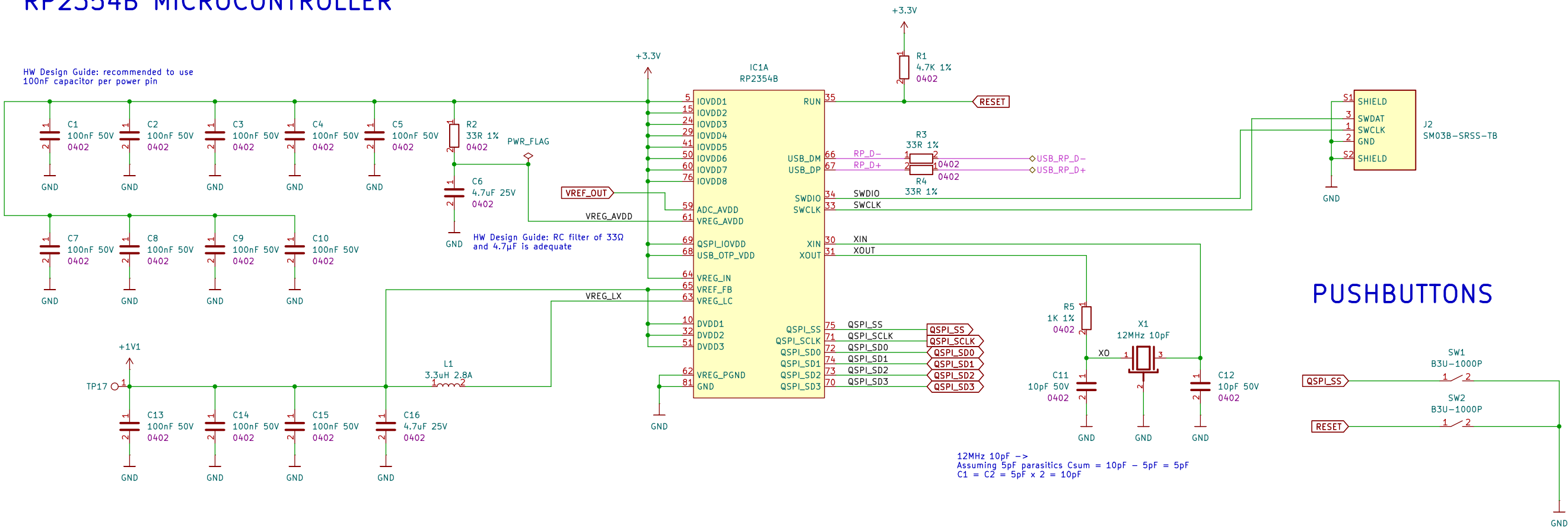
[3] Project Architecture



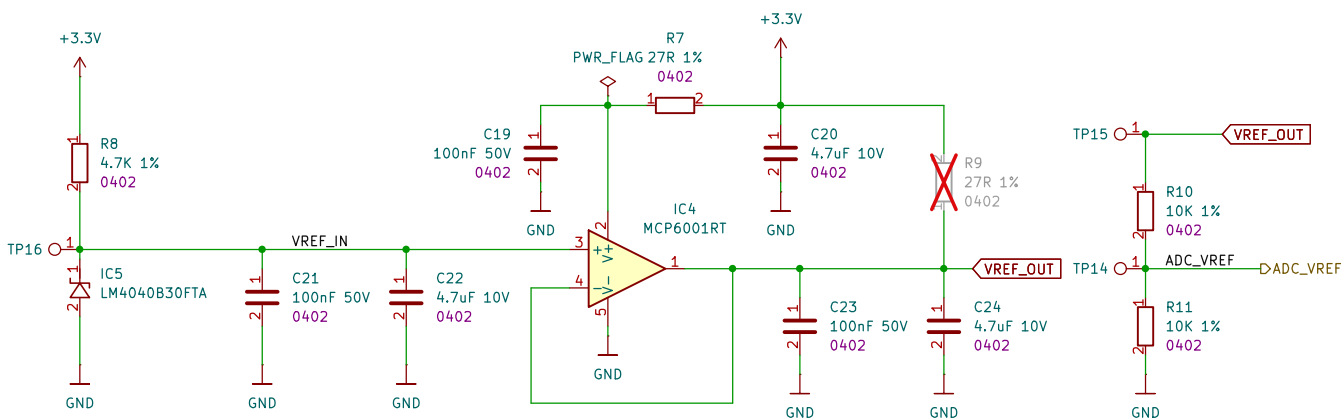
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	File Name: Project Architecture.kicad_sch		Revision: 1.0.0	Variant: C
Sheet Title: Project Architecture	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 3 of 9

[4] Microcontroller Peripherals

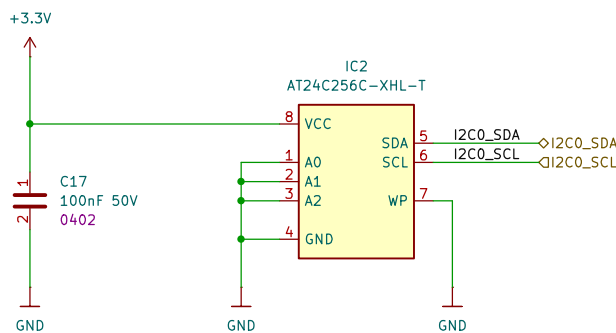
RP2354B MICROCONTROLLER



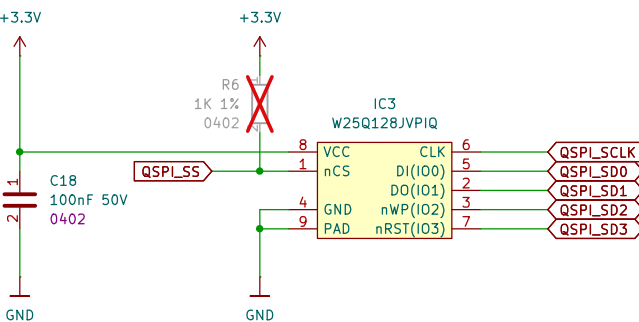
3.0V ADC VOLTAGE REFERENCE



EXTERNAL EEPROM



EXTERNAL FLASH



Date: 2026-01-31

Sheet Title:

Microcontroller Peripherals

Board Name:

BladeCore

File Name:

microcontroller_peripherals.kicad_sch

Company:

DvidMakesThings

Designer:

David Sipos

Reviewer:

Project Name:

M54

Revision:

1.0.0

Size:

A3

Variant:

C

Sheet:

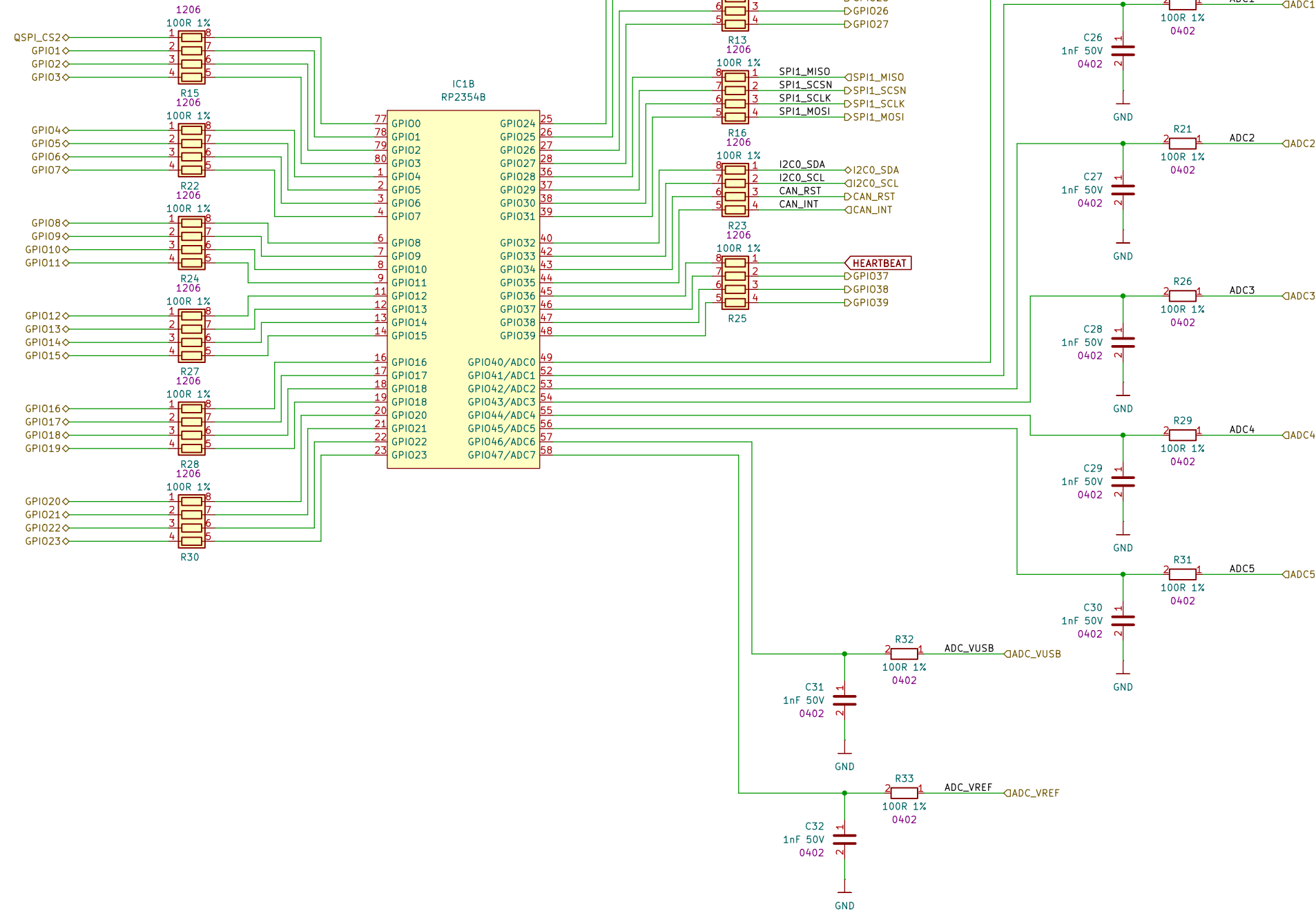
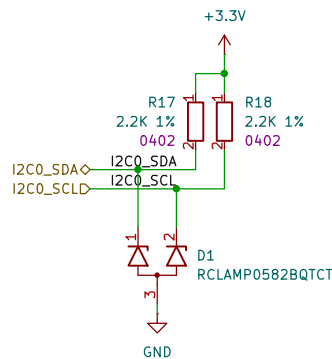
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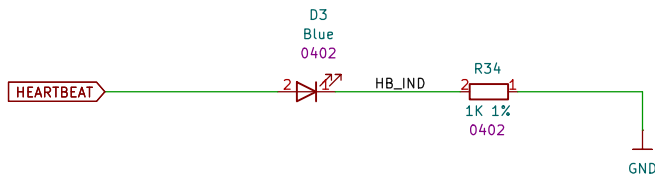
[5] Microcontroller

RP2354B GPIO ASSIGNMENT

I2C PULLUP



HEARTBEAT LED



 Date: 2026-01-31 Sheet Title: Microcontroller	Board Name: BladeCore		Project Name: M54	
	File Name: microcontroller.kicad_sch		Revision: 1.0.0	Variant: C
	Company: DvidMakesThings		Designer: David Sipos Reviewer:	Size: A3
				Sheet: 5 of 9

[6] CAN Interface

MCP2515 CAN CONTROLLER

TCAN1044 CAN TRANSCIEVER

TX0RTS/TX1RTS/TX2RTS are optional Request-to-Send inputs for the three transmit buffers (also configurable as generic digital inputs). Each has an internal ~100 k Ω pull-up to VDD, so when left unconnected the pin defaults to logic high and does not assert an RTS request.

RX0BF/RX1BF are optional receive-buffer status/interrupt outputs (also configurable as generic digital outputs). Outputs can be left open safely; no connection is required when the function is unused.

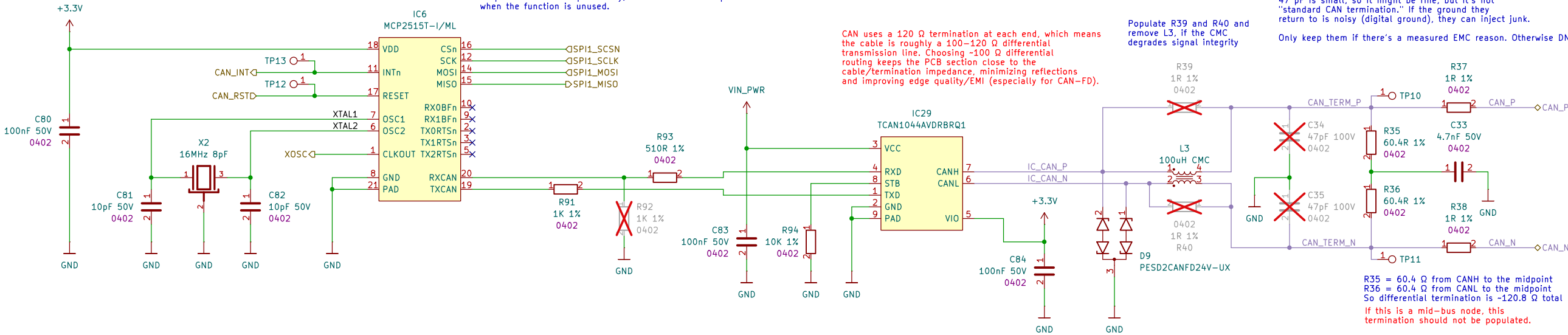
CAN uses a 120 Ω termination at each end, which means the cable is roughly a 100–120 Ω differential transmission line. Choosing ~100 Ω differential routing keeps the PCB section close to the cable/termination impedance, minimizing reflections and improving edge quality/EMI (especially for CAN-FD).

Populate R39 and R40 and remove L3, if the CMC degrades signal integrity

C34 and C35 can also be an EMC trick, but they directly load each line to ground and can mess with signal integrity depending on layout and bitrate. 47 pF is small, so it might be fine, but it's not "standard CAN termination." If the ground they return to is noisy (digital ground), they can inject junk.

Only keep them if there's a measured EMC reason. Otherwise DNI.

R35 = 60.4 Ω from CANH to the midpoint
R36 = 60.4 Ω from CANL to the midpoint
So differential termination is ~120.8 Ω total
If this is a mid-bus node, this termination should not be populated.



Date: 2026-01-31

Sheet Title:
CAN Interface

Board Name:
BladeCore

File Name:
can_interface.kicad_sch

Company:
DvidMakesThings

Designer:
David Sipos
Reviewer:

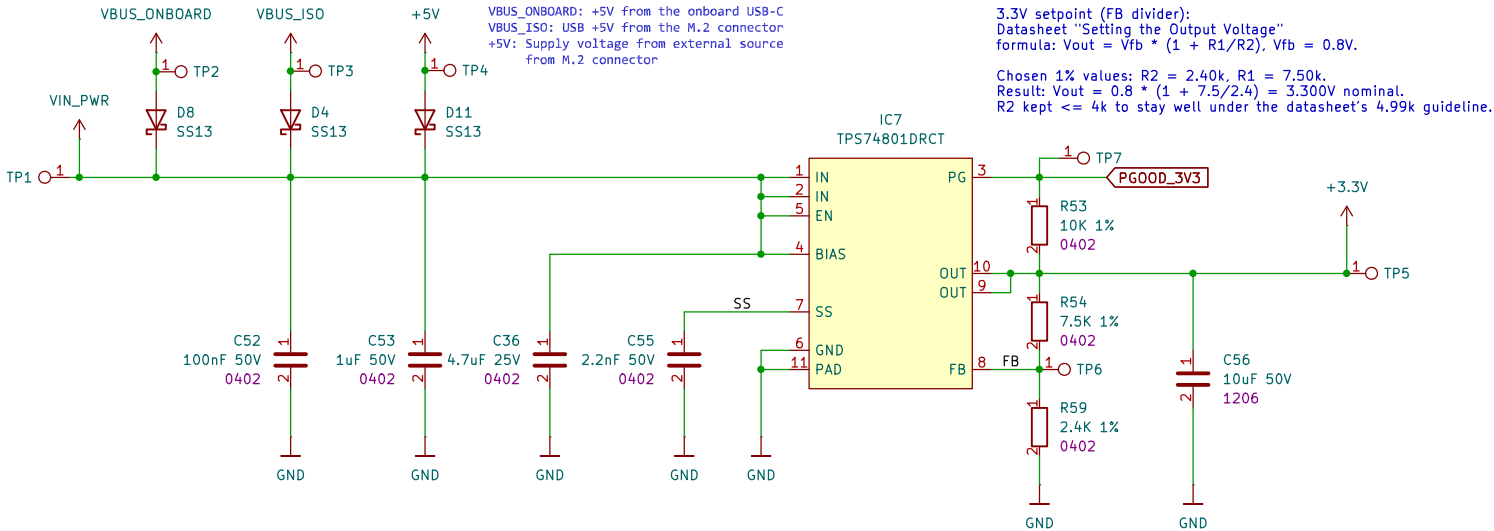
Project Name:
M54

Revision:
1.0.0
Variant:
C

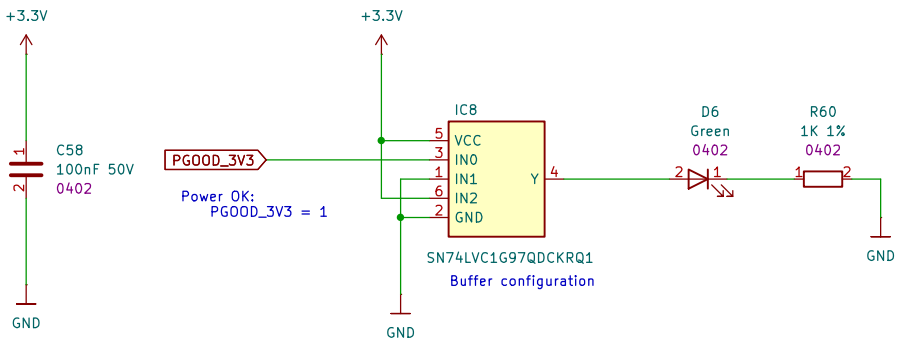
Size:
A3
Sheet:
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[7] USB Interface and Power

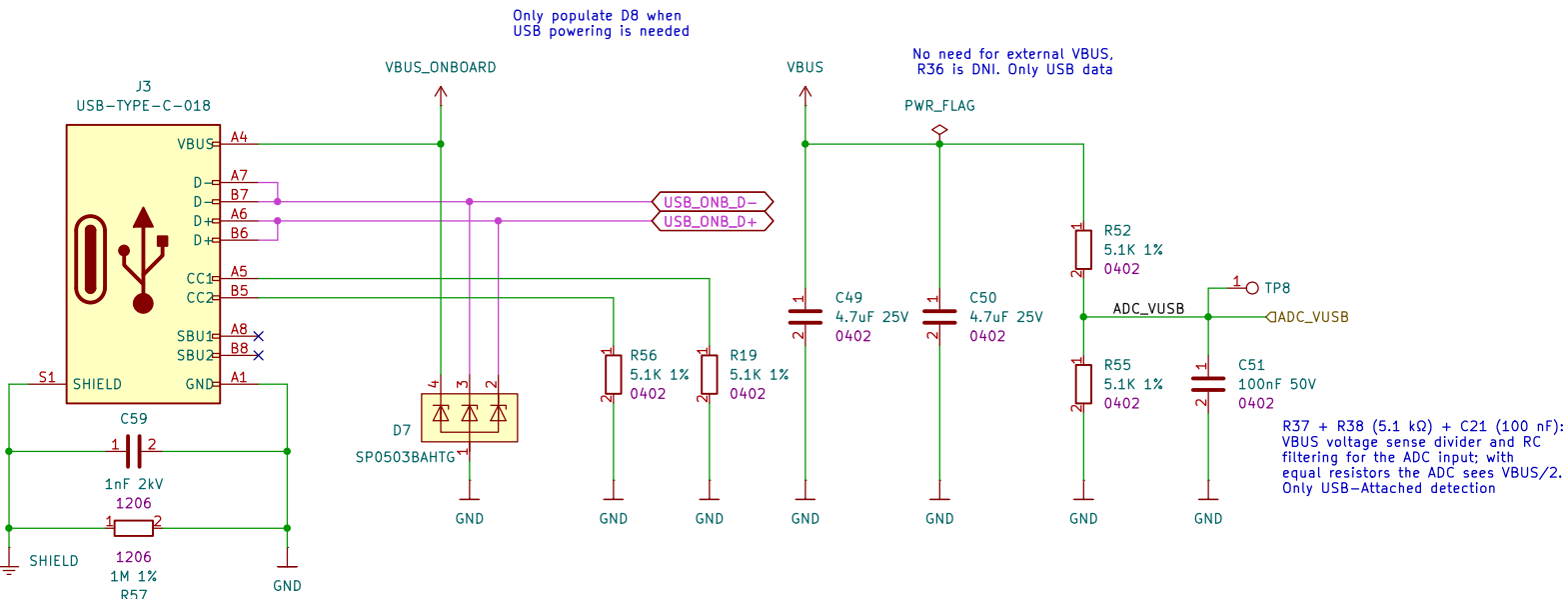
+5V to 3.3V PMIC



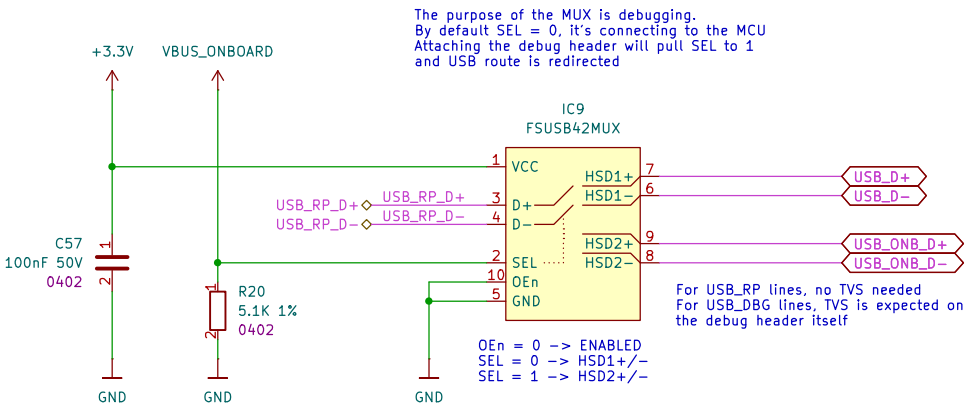
PGOOD LOGIC



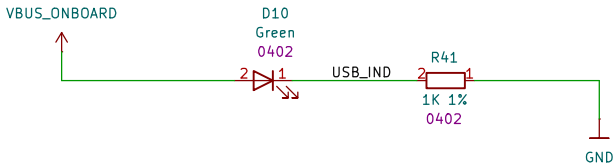
ONBOARD USB CONNECTOR



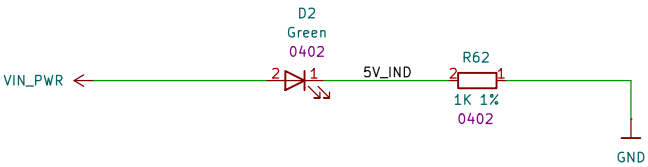
USB DATA-PATH SELECTOR



ONBOARD USB INDICATOR LED

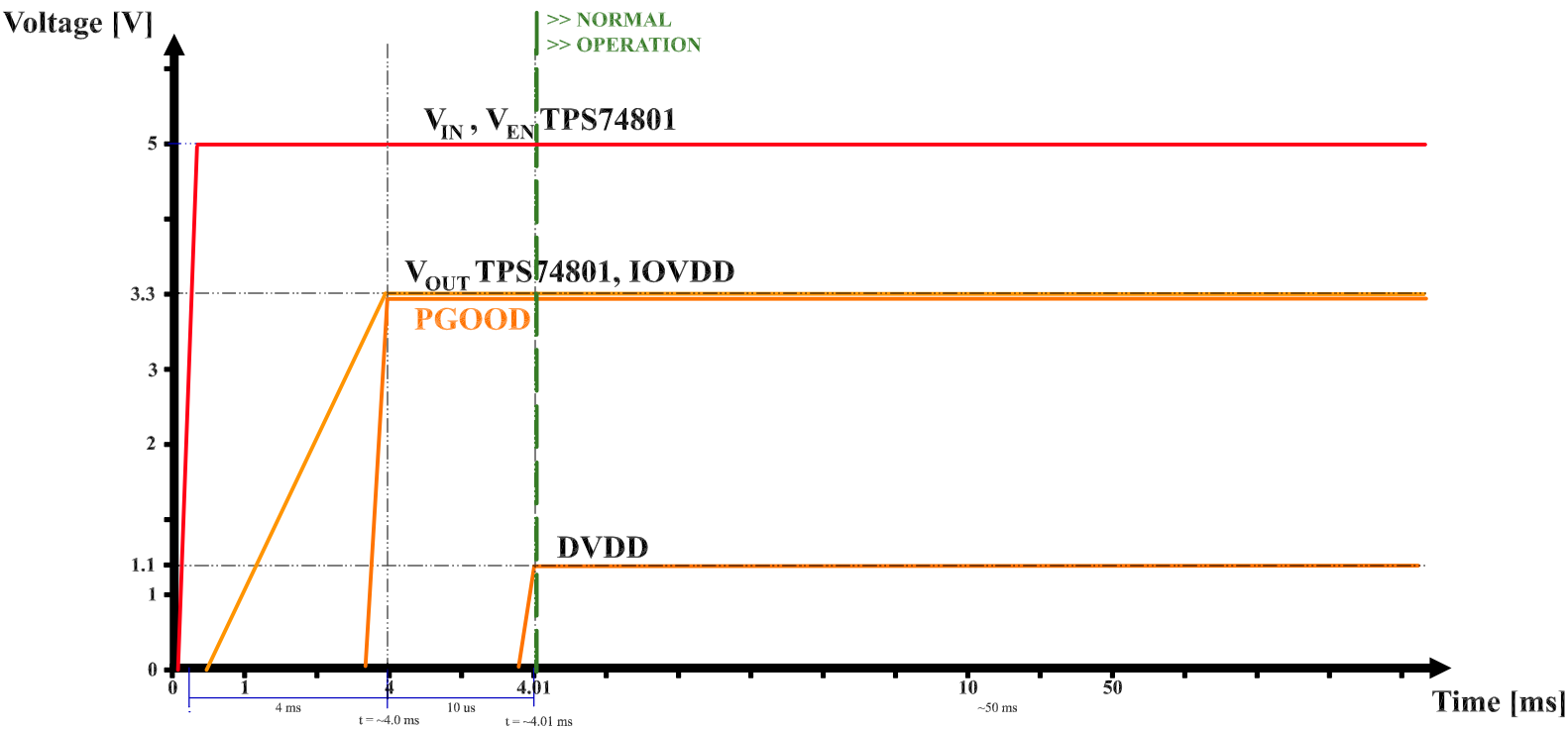


+5V INDICATOR LED



 Date: 2026-01-31 Sheet Title: USB Interface and Power	Board Name: BladeCore		Project Name: M54	
	File Name: usb_interface.kicad_sch		Revision: 1.0.0	Variant: C
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 7 of 9

[8] Power - Sequencing



#	Stage	Δt	Total elapsed
1	VIN applied (5.0V)	0	0.000 ms
2	VEN high (TPS74801 EN)	0	0.000 ms
3	VOUT = 3.3V (soft-start)	4.000 ms	4.000 ms
4	PGOOD valid	0	4.000 ms
5	RP2354B 3.3V present	0	4.000 ms
6	RP2354B 1.1V valid (POR)	0.010 ms	4.010 ms

[9] Revision History

DATE	REVISION	RESPONSIBLE	CHANGE
31.01.2026	1.0.0	DMT	INITIAL CREATION

 Date: 2026-01-31	Board Name: BladeCore		Project Name: M54	
	File Name: Revision History.kicad_sch		Revision: 1.0.0	Variant: C
	Sheet Title: Revision History	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3
				Sheet: 9 of 9